ACIDS, BASES AND INDICATORS 19

In chemistry, substances can be defined as acidic, basic or neutral. Using an indicator is one way of identifying into which of these categories a substance fits.

An **indicator** is a substance that shows the presence of a chemical, usually by changing colour.

1 Give your own definition of an indicator.

Note that sometimes alkaline solutions are mentioned. Remember that a base is the opposite of an acid. If the base is soluble, it can be referred to as an alkali.

The pH scale is often used when referring to acids and bases. A simplified version of it is shown below in figure 1.

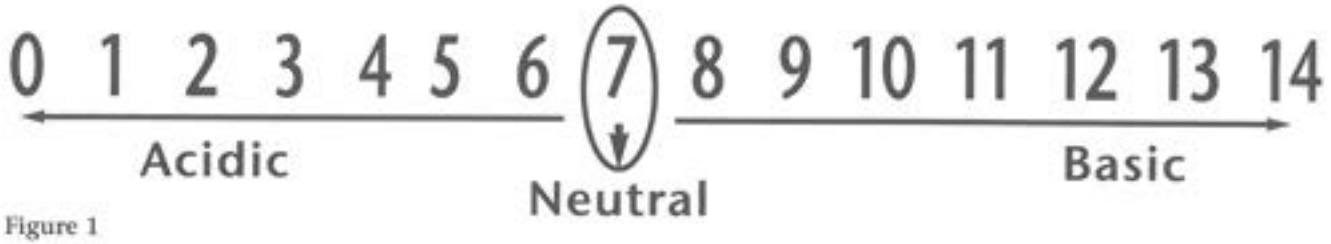


Figure 1

2 What pH number coincides with a neutral solution?

3 One solution has a pH of 2 and another solution has a pH of 4. They are both acidic. What is the pH of the more acidic solution?

4 One solution has a pH of 9 and another solution has a pH of 11. They are both basic. What is the pH of the more basic (alkaline) solution?

5 Discuss the statement: 'All alkalis are bases but not all bases are alkalis'.

The earliest indicators were made from natural sources. Some good sources of indicators are blueberries, strawberries, red cabbage, beetroots, cherries and geraniums.

6 Name two things that the sources of indicators mentioned above have in common.

Many plants that are red, blue or purple contain a class of compounds (anthocyanins) that change colour with pH. Litmus is extracted from a type of lichen and is one of the most commonly used indicators. Litmus solution has two colours: blue and red. Blue litmus turns red in the presence of an acidic solution. Red litmus turns blue in the presence of a basic solution.

7 A solution was added to blue litmus, which didn't change colour. Does this prove the solution is neutral? Explain your answer.

8 Synthetic indicators also exist; phenolphthalein is one example. What is meant by *synthetic*?

9 What properties would a synthetic indicator have to possess?

It is a common misconception that all indicators change colour at or around pH 7. Indicators can change over a wide range of pHs. Table 1 shows the pH range of some common indicators.

Indicator	pH range	Colour change
Litmus	5.0 – 8.2	Red to blue
Phenolphthalein	8.2 – 10	Colourless to pink
Turmeric	7.4 – 8.6	Yellow to red
Methyl orange	3.1 – 4.4	Red to yellow

Table 1

10 For each indicator, state the resulting colour when added to a pH 4 solution.

- **a** litmus
- **b** phenolphthalein
- **c** turmeric

11 Which of the indicators in table 1 changes colour over the widest pH range?

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